A WEIGHTING VANISHING ON A DENSE SUBSET VANISHES IDENTICALLY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 22585-22616, immediately following the statement at 22577-22583.

Editorial note. The proof opens with the author's suggestion 'We suggest reading Lemma [the specialisation lemma] first'; that sentence is retained verbatim.

Editorial note. Two cross-chapter pointers (Morphisms, Topology) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Level sets of a weighting are locally constructible. chapter lemma lemma-jumps-w, cited at proof line 22604. Makes the zero locus of the pulled-back weighting a constructible set, so that density upgrades to containing a dense open.
- The specialisation lemma for weightings. chapter lemma lemma-weighting-specialization, suggested at proof line 22586 and used through the argument. Lets vanishing be transported along specialisations, which is the mechanism behind the argument.
- A constructible dense subset contains a dense open. Topology, lemma topology-lemma-dense-in-constructible, cited at proof line 22606. Turns density plus constructibility of the zero locus into density of an open piece.
- Image of a nowhere dense set under a finite morphism. Morphisms, lemma morphisms-lemma-image-nowhere-dense-finite, cited at proof lines 22610-22611. Pushes the nowhere dense closed set Z down to a nowhere dense set in the base, so the integral vanishes on a dense open.

Source reference. Original definition 0F3A: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22152–22152. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02LK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11924–11924. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F3G: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22514–22514. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma lemma-weighting-specialization: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22469–22469. Navigation

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HX: focused-seeds/geometry/S9-morph-weighting-dense-set/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference topology-lemma-dense-in-constructible: focused-seeds/geometry/S9-morph-weighting-dense-set/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a locally quasi-finite morphism of schemes. Let $w : X \to \mathbf{Z}$ be a weighting of f. Let $E \subset X$ be a dense subset. Assume $w(x) = 0$ for all $x \in E$. Then $w = 0$.*

Proof. We suggest reading Lemma lemma-weighting-specialization first. Let $x \in X$ with image $y \in Y$. We will show that $w(x) = 0$. By Lemma 02LK we can choose a diagram

$$\begin{array}{ccccc} U & \longrightarrow & X_V & \longrightarrow & X \\ & \searrow^{\pi} & \downarrow & & \downarrow f \\ & & V & \xrightarrow{\varphi} & Y \end{array}$$

with $V \to Y$ étale, with a point $v \in V$ mapping to y, with $U \subset Y_V$ open, such that $U \to V$ is finite, and such that there exists an unique point $u \in U$ mapping to v which moreover maps to x in X. We may and do assume that V (and hence U) is affine. By Definition 0F3A the function $\int_\pi (w \circ h)$ is locally constant where $h : U \to X$ is the composition of the top horizontal arrows. The set of points U_0 where $w \circ h$ is 0 is a constructible subset of U by Lemma 0F3G and dense in U because it contains $h^{-1}(E)$ (use that h is open). By Topology, Lemma topology-lemma-dense-in-constructible we see that U_0 contains a dense open, i.e., the set of points where $w \circ h$ is nonzero is contained in a nowhere dense closed subset $Z \subset U$. Then $\pi(Z)$ is a nowhere dense closed subset of V by Morphisms, Lemma 03HX. Hence $\int_\pi (w \circ h)$ is zero on a dense open hence vanishes everywhere. However, the value at v is $w(x)[\kappa(u) : \kappa(v)]_s$ and we conclude. □